

Xiaochuan Li

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RESEARCH INTERESTS

Formal and experimental semantics; psycholinguistics; language processing and learning; computational semantics; language-model representation, evaluation, and intervention. Current case study: epistemic modality and discourse alternatives.

EDUCATION

M.Sc., Human and Artificial Intelligence 2025 - 2027

University of Technology Nuremberg (UTN)

Expected graduation: Summer 2027. Coursework: cognitive psychology, experimental design and statistics, interpretability, deep learning, epistemology.

Student · Preparatory Studies 2024 - 2025

University of Marburg · enrolled student; intensive German for graduate study

Comparing German with Chinese and English made grammatical differences-case, word order, and clause structure-concrete rather than abstract.

B.A., Philosophy 2020 - 2024

Shenzhen University

Senior thesis on Leibniz's *characteristica universalis* (advisor: Zang Yong). Coursework in philosophy of language, analytic philosophy, logic, and mathematical logic.

MANUSCRIPTS AND RESEARCH PROJECTS

Three Ways of Leaving p Unsettled - analysis in development

A formal semantic and pragmatic comparison of might p, might p or might not p, and I don't know whether p, examining the commitments they create and the responses they license. Independent judgments on the diagnostic materials are pending.

M.Sc. thesis: Epistemic possibilities in use - scope under discussion

A proposed adult behavioural study would test whether the linguistic contrasts affect later access to a specific possibility and its updating when new evidence arrives. The primary focus and any computational component remain under discussion; data collection has not begun.

Cross-Lingual Prototypicality Bias in Multimodal Evaluation Metrics and VLM Judges - first-author; in revision for arXiv

Controlled cross-lingual evaluation across seven languages and two VLM families; about 12,600 judgments, item-clustered analysis, a human control, and public code. GitHub

RESEARCH EXPERIENCE

Latent Control States - ongoing collaboration with DFKI

Designed controlled dilemma comparisons separating outcome sensitivity, framing, and action preferences. Behavioural pilots are in place; mechanistic analysis is in progress.

METHODS AND TECHNICAL SKILLS

Programming and data. Python, NumPy, pandas, scikit-learn, Git, LaTeX.

ML / NLP. PyTorch, Hugging Face Transformers, model inference, dataset pipelines, LoRA fine-tuning.

Interpretability. Linear and structural probing, activation patching, steering, ablation (working proficiency).

Experimental design and statistics. Controlled stimuli, minimal pairs, preregistration, mixed-effects analysis (working proficiency).

LANGUAGES

Mandarin Chinese (native), German (C1), English (C1), Japanese (learning).